

MEDFORD, OR, USA

WMO: 725970

Lat: 42.3769N

Lon: 122.8822W

Elev: 399

StdP: 96.62

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.5	-2.9	-8.6	1.9	2.9	-6.3	2.3	3.3	11.3	11.6	9.4	12.5	0.5	0	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.4	37.2	19.3	35.3	18.6	33.5	18.0	20.4	34.4	19.6	33.0	18.8	31.6	3.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.6	11.6	23.3	14.5	10.8	23.2	13.6	10.2	22.7	59.9	34.3	57.0	33.0	54.4	31.4	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	6.9	5.5	-7.5	40.6	2.9	1.8	-9.6	41.9	-11.3	43.0	-13.0	44.0	-15.1	45.3		
(5)			-8.0	22.1	2.8	0.9	-10.1	22.7	-11.8	23.2	-13.3	23.7	-15.4	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	4.7	6.7	9.0	11.5	15.8	19.5	24.3	23.6	19.8	13.2	7.5	4.1
	DBStd	7.69	3.04	2.97	3.15	3.66	3.93	3.76	3.13	2.97	3.59	3.42	3.26	3.54
	HDD10.0	629	165	98	58	24	2	0	0	0	0	9	88	185
	HDD18.3	2331	422	326	289	206	100	30	1	2	27	162	326	441
	CDD10.0	1852	2	6	27	70	181	284	443	422	294	109	12	2
	CDD18.3	512	0	0	0	2	21	64	186	166	71	4	0	0
	CDH23.3	6986	0	0	4	83	399	941	2375	2061	1008	114	0	0
	CDH26.7	3527	0	0	0	21	153	423	1313	1105	485	28	0	0
Wind	WSAvg	1.7	1.3	1.6	1.9	2.0	2.2	2.3	2.2	1.9	1.6	1.3	1.2	1.3
Precipitation	PrecAvg	476	68	50	48	33	29	16	6	11	17	36	76	87
	PrecMax	798	157	144	112	91	108	66	41	72	107	159	195	323
	PrecMin	229	5	5	7	6	0	0	0	0	0	0	11	9
	PrecStd	116	40	33	26	20	21	14	10	16	21	32	47	64
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	19.3	23.7	29.5	34.1	37.2	40.0	39.7	36.8	30.0	19.9	15.7
		MCWB	9.7	11.4	13.0	15.3	17.6	19.7	20.3	19.6	17.7	15.6	12.0	10.2
	2%	DB	13.5	16.5	20.8	25.7	30.8	33.9	37.6	36.7	34.4	26.5	17.0	13.2
		MCWB	8.7	9.8	11.6	14.3	16.8	18.4	19.9	18.8	17.3	14.3	11.1	8.9
	5%	DB	11.3	14.4	18.2	22.5	27.8	31.4	35.6	34.7	32.0	23.7	14.8	11.1
		MCWB	7.9	8.7	10.7	12.7	15.4	17.2	19.2	18.3	16.8	13.4	10.2	8.5
	10%	DB	9.8	12.4	15.7	19.3	24.9	28.8	33.5	32.8	29.5	21.0	12.8	9.3
		MCWB	7.2	7.9	9.7	11.4	14.3	16.3	18.6	17.7	15.9	12.5	9.6	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	12.6	14.0	16.1	19.0	20.7	22.0	20.8	19.6	16.6	14.2	11.7
		MCDB	13.2	17.8	21.8	27.1	31.4	34.4	36.4	35.8	32.4	26.8	17.9	13.6
	2%	WB	9.8	10.9	12.5	14.7	17.6	19.3	20.8	19.8	18.3	15.4	12.4	10.0
		MCDB	12.5	14.6	18.5	24.3	28.6	31.9	35.0	34.2	30.8	24.1	15.2	12.2
	5%	WB	8.6	9.7	11.4	13.3	16.3	17.9	19.9	19.1	17.4	14.2	11.2	8.7
		MCDB	10.7	13.3	16.8	21.1	25.9	29.7	33.5	32.6	29.7	21.4	13.7	10.6
	10%	WB	7.6	8.5	10.2	11.9	15.0	16.8	19.1	18.3	16.5	13.1	10.0	7.6
		MCDB	9.6	11.7	14.9	18.3	23.4	27.4	32.0	30.9	28.1	19.3	12.5	9.3
Mean Daily Temperature Range	5% DB	MDBR	8.1	10.4	11.7	12.6	14.1	15.5	17.4	17.6	17.4	14.4	9.0	7.1
		MCDBR	11.1	14.2	17.2	19.3	19.8	19.8	20.0	20.1	21.7	19.7	12.9	9.4
		MCWBR	6.9	8.3	9.0	9.0	7.9	7.2	6.8	6.3	7.7	8.9	7.5	5.8
	5% WB	MCDBR	8.9	12.0	14.9	17.5	18.0	18.7	18.6	18.7	19.7	16.8	10.2	7.6
		MCWBR	6.0	7.3	7.9	8.5	7.4	7.0	6.6	6.2	7.4	8.0	6.3	5.0
Clear-Sky Solar Irradiance	taub	0.305	0.307	0.310	0.332	0.340	0.337	0.343	0.340	0.320	0.317	0.302	0.311	
	taud	2.551	2.559	2.571	2.486	2.498	2.530	2.527	2.524	2.562	2.578	2.587	2.527	
	Ebn at Noon	848	906	942	937	933	933	924	918	918	881	847	804	
	Edh at Noon	69	80	88	104	105	102	102	99	88	77	66	65	
All-Sky Solar Radiation	RadAvg	1.72	2.58	3.70	5.10	6.51	7.54	7.95	6.93	5.38	3.44	1.93	1.38	
	RadStd	0.26	0.36	0.39	0.49	0.54	0.45	0.28	0.33	0.31	0.34	0.24	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+70
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+4

Nomenclature: See separate page